

Lesson 7-1: Explore Integers and Their Opposites

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.5

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

On a number line marked in ones, what number is three steps right of 2?

Answer: 5

Count 3, 4, 5 to the right.

Prerequisite skill: Place whole numbers on a number line

TASK 2 • FOUNDATION 2

Start at 2 and move backward 5 steps. Where do you land?

Answer: -3

Count 1, 0, -1, -2, -3.

Prerequisite skill: Count forward and backward through zero

TASK 3 • FOUNDATION 3

Write the integer for 7 degrees below zero.

Answer: -7

The negative sign indicates a value below zero.

Prerequisite skill: Read a negative sign as “below zero” or “the opposite of”

TASK 4 • FOUNDATION 4

Simplify $3x + 4x$.

Answer: $7x$

Add the coefficients of the like terms.

Prerequisite skill: Simplify an algebraic expression carried in from Unit 6

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Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

What is the opposite of -6 ? Where does -3 sit compared with 0?

Answer: Opposite of -6 is $+6$ (or 6). Position of -3 : sits 3 units to the left of 0 (or 3 units below 0).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

What is the opposite of -9 ? Where does -5 sit compared with 0?

Answer: The opposite of -9 is 9. The number -5 sits 5 units to the left of 0.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

What is the opposite of 12? Where does -1 sit compared with 0?

Answer: The opposite of 12 is -12 . The number -1 sits 1 unit to the left of 0.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A diver is at -18 feet and a bird is at the opposite elevation. How far apart are they?

Answer: The bird is at $+18$ feet. The distance is $18 + 18 = 36$ feet.

Check the result against the situation and explain your reasoning.

Lesson 7-2: Represent Rational Numbers and Their Opposites on the Number Line

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Write the decimal halfway between 1 and 2.

Answer: 1.5

Half of a unit is 0.5; add it to 1.

Prerequisite skill: Place fractions and decimals between whole numbers

TASK 2 • FOUNDATION 2

Between which two consecutive integers is -2.6?

Answer: -3 and -2

-2.6 lies to the right of -3 and to the left of -2.

Prerequisite skill: Name the two consecutive integers a number falls between

TASK 3 • FOUNDATION 3

Write $\frac{3}{5}$ as a decimal.

Answer: 0.6

Convert $\frac{3}{5}$ to $\frac{6}{10}$, or divide 3 by 5.

Prerequisite skill: Convert between a fraction and a decimal

TASK 4 • FOUNDATION 4

An interval from 0 to 1 is split into 4 equal parts. What is the length of one part?

Answer: $\frac{1}{4}$

Divide the unit length into four equal shares.

Prerequisite skill: Partition a number-line interval into equal parts

Lesson 7-2: Represent Rational Numbers and Their Opposites on the Number Line

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Which two integers does 2.7 fall between? Where does $-11/2$ go?

Answer: 2.7 falls between integers 2 and 3. $-11/2 = -5.5$, which falls between -6 and -5 (exactly halfway between them).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Which two integers does 3.4 fall between? Where does $-7/2$ go?

Answer: 3.4 falls between 3 and 4. $-7/2 = -3.5$, which sits halfway between -4 and -3 .

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Which two integers does -1.2 fall between? Where does $9/4$ go?

Answer: -1.2 falls between -2 and -1 . $9/4 = 2.25$, which sits between 2 and 3, a quarter of the way up.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Place $-2 \frac{1}{4}$, -2.5 , and -2.75 in order from least to greatest.

Answer: -2.75 , -2.5 , $-2 \frac{1}{4}$ (which is -2.25). Further left means less.

Check the result against the situation and explain your reasoning.

Lesson 7-3: Understand Absolute Value of Rational Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.7.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

What is the distance from -4 to 0 ?

Answer: 4

Count four unit intervals to zero.

Prerequisite skill: Count the distance between two points on a number line

TASK 2 • FOUNDATION 2

What is the opposite of -7 ?

Answer: 7

Opposites are equally far from zero on opposite sides.

Prerequisite skill: Name the opposite of a number

TASK 3 • FOUNDATION 3

Evaluate the absolute value of -9 .

Answer: 9

Absolute value measures distance from zero, which is 9.

Prerequisite skill: Report a distance as a positive number

TASK 4 • FOUNDATION 4

A path goes 3 units to zero, then 5 more units. What total distance is traveled?

Answer: 8

Add the two nonnegative distances: $3 + 5$.

Prerequisite skill: Add two distances to find a total

Lesson 7-3: Understand Absolute Value of Rational Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.7.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

$|-9|$? How far is -4 from 0 ? How far from -4 to 3 ?

Answer: $|-9| = 9$. Distance from -4 to 0 is **4 units**. Distance from -4 to 3 : **4 units (to zero) + 3 units (to 3) = 7 units total**.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

$|-12|$? How far is -7 from 0 ? How far from -7 to 2 ?

Answer: $|-12| = 12$. Distance from -7 to 0 is **7**. From -7 to 2 : **$7 + 2 = 9$ units**.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

$|-3.5|$? How far is -6 from 0 ? How far from -2 to 8 ?

Answer: $|-3.5| = 3.5$. Distance from -6 to 0 is **6**. From -2 to 8 : **$2 + 8 = 10$ units**.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Two numbers both have absolute value 6.5 . What are they, and how far apart are they?

Answer: **6.5 and -6.5 . They are 13 units apart.**

Check the result against the situation and explain your reasoning.

Lesson 7-4: Compare and Order Integers and Rational Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.7.a

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Fill the blank with $<$ or $>$: -8 ___ -3 .

Answer: $<$

-8 is farther left on the number line, so it is less.

Prerequisite skill: Use $<$ and $>$ correctly

TASK 2 • FOUNDATION 2

Order -1 , 3 , -4 , 0 from least to greatest.

Answer: -4 , -1 , 0 , 3

Read positions on the number line from left to right.

Prerequisite skill: Place numbers on a number line

TASK 3 • FOUNDATION 3

Which is greater: $\frac{3}{4}$ or 0.7 ?

Answer: $\frac{3}{4}$

Convert $\frac{3}{4}$ to 0.75 ; 0.75 is greater than 0.70 .

Prerequisite skill: Compare decimals and fractions

TASK 4 • FOUNDATION 4

Point A is at -5 and point B is at -2 . Which point is farther right?

Answer: B

-2 is to the right of -5 and is therefore greater.

Prerequisite skill: Read left-is-less, right-is-greater on a number line

Lesson 7-4: Compare and Order Integers and Rational Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.7.a

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

-6 ____ -2 . Order these: -3 , 1.5 , $-1/2$, 0 .

Answer: $-6 < -2$ (since -6 is further to the left on the number line). Ordered least to greatest: -3 , $-1/2$, 0 , 1.5 .

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

-8 ____ -3 . Order these: -5 , 2.5 , $-1/2$, 0 .

Answer: $-8 < -3$. Order: -5 , $-1/2$, 0 , 2.5 .

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

-1 ____ -7 . Order these: -2 , 0.75 , $-3/4$, 3 .

Answer: $-1 > -7$. Order: -2 , $-3/4$, 0.75 , 3 .

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A temperature of -4°F rises 6 degrees. Is it now warmer or colder than -1°F ?

Answer: $-4 + 6 = 2^{\circ}\text{F}$, which is warmer than -1°F because 2 sits to the right of -1 .

Check the result against the situation and explain your reasoning.

Lesson 7-5: Represent Rational Numbers on the Coordinate Plane

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.b

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

From the origin, move 2 right and 6 up. Give the ordered pair.

Answer: (2, 6)

The x-coordinate records horizontal movement; y records vertical movement.

Prerequisite skill: Plot ordered pairs in Quadrant I

TASK 2 • FOUNDATION 2

Which quadrant contains $(-3, 4)$?

Answer: Quadrant II

Negative x is left and positive y is up.

Prerequisite skill: Read the sign of each coordinate to name a quadrant

TASK 3 • FOUNDATION 3

Describe the moves from the origin to $(-4, -2)$.

Answer: 4 left, 2 down

Move horizontally first; both signs are negative.

Prerequisite skill: Move left or right first, then up or down from the origin

TASK 4 • FOUNDATION 4

An axis counts by 5. What value is two tick intervals left of zero?

Answer: -10

Two intervals total 10 units; left of zero is negative.

Prerequisite skill: Read a scaled axis before plotting

Lesson 7-5: Represent Rational Numbers on the Coordinate Plane

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.b

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Which quadrant holds $(-4, 5)$? Plot $(3, -2)$.

Answer: $(-4, 5)$ has negative x and positive y , which is Quadrant II. Plot $(3, -2)$: Move 3 units right on x -axis, then 2 units down on y -axis (Quadrant IV).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Which quadrant holds $(-3, -6)$? Plot $(5, -1)$.

Answer: $(-3, -6)$ has both coordinates negative — Quadrant III. $(5, -1)$: 5 right, 1 down — Quadrant IV.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Which quadrant holds $(7, 2)$? Plot $(-2, 4)$.

Answer: $(7, 2)$ has both coordinates positive — Quadrant I. $(-2, 4)$: 2 left, 4 up — Quadrant II.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Reflect $(-3, 4)$ across the y -axis, then across the x -axis. Where does it end up?

Answer: Across the y -axis: $(3, 4)$. Then across the x -axis: $(3, -4)$.

Check the result against the situation and explain your reasoning.

Lesson 7-6: Determine Distance on the Coordinate Plane

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.8

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Compute $6 - (-2)$.

Answer: 8

The distance from -2 to 6 is $2 + 6 = 8$.

Prerequisite skill: Subtract integers on a number line

TASK 2 • FOUNDATION 2

Evaluate the absolute value of -12.

Answer: 12

Distance from zero is nonnegative.

Prerequisite skill: Use absolute value to report a distance

TASK 3 • FOUNDATION 3

A plotted point is 3 units left and 5 units up from the origin. Give its coordinates.

Answer: (-3, 5)

Left gives negative x; up gives positive y.

Prerequisite skill: Read the coordinates off a plotted point

TASK 4 • FOUNDATION 4

Find the horizontal distance from $(-4, 2)$ to $(7, 2)$.

Answer: 11

The segment crosses the y-axis: 4 units to zero plus 7 more units.

Prerequisite skill: Add two distances when a segment crosses an axis

Lesson 7-6: Determine Distance on the Coordinate Plane

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.8

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Distance from $(-5, 2)$ to $(3, 2)$? What is $|-5 - 3|$?

Answer: Distance from $(-5, 2)$ to $(3, 2)$: Segment is horizontal ($y=2$). Distance = $|-5 - 3| = |-8| = 8$ units (or 5 units to y -axis + 3 units to right = 8).

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Distance from $(-6, 3)$ to $(2, 3)$? What is $|-6 - 2|$?

Answer: The segment is horizontal ($y = 3$). Distance = $|-6 - 2| = |-8| = 8$ units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Distance from $(4, -5)$ to $(4, 7)$? What is $|-5 - 7|$?

Answer: The segment is vertical ($x = 4$). Distance = $|-5 - 7| = |-12| = 12$ units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Points $A(-3, 5)$ and $B(-3, -2)$. Find AB , then find a point C so that $BC = 4$ horizontally.

Answer: $AB = |5 - (-2)| = 7$ units. C could be $(1, -2)$ or $(-7, -2)$.

Check the result against the situation and explain your reasoning.

Lesson 7-7: Represent Polygons on the Coordinate Plane

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Plot (0, 0), (4, 0), (4, 3), (0, 3) and connect in order. Name the shape.

Answer: Rectangle

Opposite sides are parallel and all four corners are right angles.

Prerequisite skill: Plot several points and connect them in order

TASK 2 • FOUNDATION 2

Find the vertical distance from (2, -3) to (2, 5).

Answer: 8

The x-coordinates match; distance is $3 + 5 = 8$.

Prerequisite skill: Find horizontal and vertical distances between vertices

TASK 3 • FOUNDATION 3

Find the perimeter of a rectangle with side lengths 4 and 3.

Answer: 14

Add all four sides: $4 + 3 + 4 + 3$.

Prerequisite skill: Add side lengths to find perimeter

TASK 4 • FOUNDATION 4

Find the area of a triangle with base 8 and height 5.

Answer: 20

Half of the product 8×5 is 20.

Prerequisite skill: Find the area of a rectangle or triangle

Lesson 7-7: Represent Polygons on the Coordinate Plane

Connect & apply • 8 tasks • Prepares for Standard 6.G.A.3

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Plot (1,1), (5,1), (5,4), (1,4). What is the perimeter? The area?

Answer: Vertices form a rectangle: Horizontal length: $5 - 1 = 4$ units. Vertical height: $4 - 1 = 3$ units. Perimeter: $2(4 + 3) = 14$ units. Area: $4 \times 3 = 12$ square units.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Plot (2, 2), (8, 2), (8, 6), (2, 6). What is the perimeter? The area?

Answer: Length $8 - 2 = 6$, height $6 - 2 = 4$. Perimeter = $2(6 + 4) = 20$ units. Area = $6 \times 4 = 24$ square units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Plot (−3, 1), (3, 1), (3, 5), (−3, 5). What is the perimeter? The area?

Answer: Length $3 - (-3) = 6$, height $5 - 1 = 4$. Perimeter = 20 units. Area = 24 square units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

Three vertices of a rectangle are (1, 2), (7, 2), and (7, 9). Give the fourth vertex and the perimeter.

Answer: Fourth vertex (1, 9). Sides 6 and 7, so perimeter = $2(6 + 7) = 26$ units.

Check the result against the situation and explain your reasoning.

Lesson 7-8: Ordered Pairs in All Four Quadrants

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Describe how to plot (5, 3) starting at the origin.

Answer: Move 5 units right along the x-axis, then 3 units up.

The first coordinate is horizontal; the second is vertical.

Prerequisite skill: Plot and read a point in Quadrant I: across first, then up

TASK 2 • FOUNDATION 2

What integer is 6 units to the left of 2 on a number line?

Answer: -4

Count left from 2: $2 - 6 = -4$.

Prerequisite skill: Locate integers on horizontal and vertical number lines

TASK 3 • FOUNDATION 3

How far apart are the points (–3, 5) and (4, 5)?

Answer: 7

Same y-coordinate: 3 units to the y-axis plus 4 units beyond, so 7 units.

Prerequisite skill: Find the distance between two points that share a coordinate

TASK 4 • FOUNDATION 4

A rectangle is 6 units by 3 units. What is its perimeter?

Answer: 18

$2 \times 6 + 2 \times 3 = 12 + 6 = 18$ units.

Prerequisite skill: Find the perimeter and area of a rectangle

Lesson 7-8: Ordered Pairs in All Four Quadrants

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.c

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

What integer is 5 units left of 3? Describe how to reach (4, 6) from the origin. Area of a 7-by-2 rectangle?

Answer: $3 - 5 = -2$. Move 4 units right, then 6 units up. $7 \times 2 = 14$ square units.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

What integer is 7 units left of 4? Describe how to reach (2, 8) from the origin. Area of a 5-by-3 rectangle?

Answer: -3 . Move 2 units right, then 8 units up. 15 square units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

What integer is 4 units below 1? Describe how to reach (6, 1) from the origin. Perimeter of a 6-by-4 rectangle?

Answer: -3 . Move 6 units right, then 1 unit up. $2 \times 6 + 2 \times 4 = 20$ units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A rectangle has corners $(-6, -3)$, $(2, -3)$, $(2, 1)$ and $(-6, 1)$. Find its perimeter.

Answer: The sides are $6 + 2 = 8$ and $3 + 1 = 4$ units, so the perimeter is $2 \times 8 + 2 \times 4 = 24$ units.

Check the result against the situation and explain your reasoning.

Lesson 7-9: Reflect Points Across Axes

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.b

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 1 • FOUNDATION 1

Which quadrant holds $(-6, -1)$? Explain using the signs.

Answer: Quadrant III, because both coordinates are negative.

Left of the y-axis and below the x-axis is Quadrant III.

Prerequisite skill: Name the quadrant of a point from the signs of its coordinates

TASK 2 • FOUNDATION 2

What is the opposite of -9 ?

Answer: 9

-9 is 9 units left of 0; its opposite is 9 units right: 9.

Prerequisite skill: Find the opposite of an integer

TASK 3 • FOUNDATION 3

Is the point $(0, -5)$ on the x-axis or the y-axis? Explain.

Answer: The y-axis, because its x-coordinate is 0.

A point with $x = 0$ has not moved left or right.

Prerequisite skill: Recognize points on an axis by a zero coordinate

TASK 4 • FOUNDATION 4

How far is the point $(-7, 3)$ from the y-axis?

Answer: 7

The distance is $|-7| = 7$ units.

Prerequisite skill: Find a point's distance from an axis using absolute value

Lesson 7-9: Reflect Points Across Axes

Connect & apply • 8 tasks • Prepares for Standard 6.NS.C.6.b

Facilitation note: Accept equivalent values unless the question requests a specific representation, scale, or simplest form. Discuss units and reasoning.

TASK 5 • QUICK CHECK A

Opposite of -6 ? Quadrant of $(4, -2)$? Distance from $(-5, 1)$ to the y -axis?

Answer: The opposite of -6 is 6. Quadrant IV: x positive, y negative. $|-5| = 5$ units.

Check every part of the prompt and explain why your method works.

TASK 6 • QUICK CHECK B

Opposite of 8? Quadrant of $(-3, -7)$? Distance from $(2, -9)$ to the x -axis?

Answer: -8 . Quadrant III. $|-9| = 9$ units.

Check every part of the prompt and explain why your method works.

TASK 7 • QUICK CHECK C

Opposite of -11 ? Quadrant of $(-1, 5)$? Distance from $(6, 4)$ to the y -axis?

Answer: 11. Quadrant II. $|6| = 6$ units.

Check every part of the prompt and explain why your method works.

TASK 8 • TRANSFER & EXTEND

A triangle has vertices $(1, 2)$, $(4, 2)$ and $(1, 6)$. Reflect it across the y -axis and give the image's vertices.

Answer: $(-1, 2)$, $(-4, 2)$ and $(-1, 6)$.

Check the result against the situation and explain your reasoning.